

200 Terraform Interview Q&A

1. What is Terraform?

Terraform is an open-source infrastructure as code (IaC) tool developed by HashiCorp that allows users to define and provision data center infrastructure using a high-level configuration language called HCL (HashiCorp Configuration Language) or JSON.

Example:

```
provider "aws" {
  region = "us-west-2"
}

resource "aws_instance" "example" {
  ami          = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
}
```

2. What are the main features of Terraform?

Terraform's main features include Infrastructure as Code (IaC), execution plans, resource graphs, change automation, and state management.

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}
```

3. What is the difference between Terraform and other IaC tools like Ansible, Puppet, and Chef?

Terraform focuses on infrastructure provisioning, is declarative, and uses HCL. Tools like Ansible, Puppet, and Chef focus on configuration management and are procedural.

Example:

```
resource "aws_instance" "example" {
  ami          = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
}
```

4. What is a provider in Terraform?

A provider is a plugin that Terraform uses to manage an external API. Providers define the resources and data sources available.

Example:

```
provider "aws" {
  region = "us-west-2"
}
```

5. How does Terraform manage dependencies?

Terraform uses a dependency graph to manage dependencies between resources. It automatically understands the order of operations needed based on resource dependencies.

Example:

```
resource "aws_instance" "web" {
  ami          = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
  subnet_id    = aws_subnet.example.id
}

resource "aws_subnet" "example" {
  vpc_id        = aws_vpc.example.id
  cidr_block    = "10.0.1.0/24"
}

resource "aws_vpc" "example" {
  cidr_block = "10.0.0.0/16"
}
```

6. What is a state file in Terraform?

A state file is a file that Terraform uses to keep track of the current state of the infrastructure. It maps the resources defined in the configuration to the real-world resources.

Example:

```
terraform show
```

7. Why is it important to manage the state file in Terraform?

Managing the state file is crucial because it ensures consistency between the infrastructure's real state and the configuration. It also enables features like change detection and planning.

Example:

```
terraform init
```

8. How can you secure the state file in Terraform?

State files can be secured by storing them in remote backends with proper access controls and encryption, such as AWS S3 with server-side encryption and access control policies.

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
    encrypt = true
  }
}
```

9. What are modules in Terraform?

Modules are reusable packages of Terraform configurations that can be shared and composed to manage resources efficiently.

Example:

```
module "vpc" {
  source = "../modules/vpc"
}
```

10. What is the purpose of the `terraform init` command?

`terraform init` initializes a working directory containing Terraform configuration files, downloads the necessary provider plugins, and prepares the environment.

Example:

```
terraform init
```

11. What does the `terraform plan` command do?

`terraform plan` creates an execution plan, showing what actions Terraform will take to achieve the desired state defined in the configuration.

Example:

```
terraform plan
```

12. What is the `terraform apply` command used for?

`terraform apply` applies the changes required to reach the desired state of the configuration. It executes the plan created by `terraform plan`.

Example:

```
terraform apply
```

13. What is the purpose of the `terraform destroy` command?

`terraform destroy` is used to destroy the infrastructure managed by Terraform. It removes all the resources defined in the configuration.

Example:

```
terraform destroy
```

14. How do you define and use variables in Terraform?

Variables in Terraform are defined using the `variable` block and can be used by referring to them with `var.<variable_name>`.

Example:

```
variable "instance_type" {
  description = "Type of EC2 instance"
  default     = "t2.micro"
}

resource "aws_instance" "example" {
  ami           = "ami-0c55b159cbf1f0"
  instance_type = var.instance_type
}
```

15. What are output values in Terraform and how are they used?

Output values are used to extract information from the resources and make it accessible after the `apply` phase. They can be used to output resource attributes.

Example:

```
output "instance_id" {
  value = aws_instance.example.id
}
```

16. How do you manage different environments (e.g., dev, prod) in Terraform?

Different environments can be managed using workspaces or separate directories with different variable files and state files.

Example:

```
terraform workspace new dev
terraform workspace new prod
```

17. What is remote state and how do you configure it in Terraform?

Remote state allows Terraform to store the state file in a remote storage backend, enabling team collaboration and secure storage.

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}
```

18. How do you import existing resources into Terraform?

Existing resources can be imported using the `terraform import` command, which maps the existing resource to a Terraform resource in the state file.

Example:

```
terraform import aws_instance.example i-1234567890abcdef0
```

19. What are data sources in Terraform?

Data sources allow Terraform to fetch data from existing infrastructure or services to use in resource definitions.

Example:

```
data "aws_ami" "example" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn-ami-hvm-*"]
  }
}
```

20. What are provisioners in Terraform?

Provisioners are used to execute scripts or commands on a local or remote machine as part of the resource lifecycle.

Example:

```
resource "aws_instance" "example" {
  ami          = "ami-0c55b159cbfafa1f0"
  instance_type = "t2.micro"

  provisioner "local-exec" {
    command = "echo ${self.public_ip} > ip_address.txt"
  }
}
```

21. How do you handle secrets in Terraform?

Secrets can be managed using environment variables, secure secret management services (e.g., AWS Secrets Manager), or Terraform's sensitive attribute.

Example:

```
resource "aws_secretsmanager_secret" "example" {
  name          = "example"
  description   = "An example secret"
}

resource "aws_secretsmanager_secret_version" "example" {
  secret_id     = aws_secretsmanager_secret.example.id
  secret_string = jsonencode({
    username = "example_user"
    password = "example_password"
  })
}
```

22. What is a backend in Terraform?

A backend in Terraform defines where and how state is loaded and stored. It can be local or remote (e.g., S3, Consul, etc.).

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}
```

23. How do you use conditional expressions in Terraform?

Conditional expressions in Terraform are used to assign values based on conditions using the ternary operator `condition ? true_value : false_value`.

Example:

```
variable "environment" {
  default = "dev"
}

resource "aws_instance" "example" {
  ami          = "ami-0c55b159cbfaffe1f0"
  instance_type = var.environment == "prod" ? "t2.large" : "t2.micro"
}
```

24. What is the purpose of the `terraform validate` command?

`terraform validate` is used to validate the syntax and configuration of the Terraform files without creating any resources.

Example:

```
terraform validate
```

25. How can you format Terraform configuration files?

Terraform configuration files can be formatted using the `terraform fmt` command, which formats the files according to the

Terraform style guide.

Example:

```
terraform fmt
```

26. What is the difference between `count` and `for_each` in Terraform?

`count` is used to create multiple instances of a resource, while `for_each` is used to iterate over a map or set of values to create multiple instances.

Example (count):

```
resource "aws_instance" "example" {
  count      = 3
  ami       = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
}
```

Example (for_each):

```
resource "aws_instance" "example" {
  for_each   = toset(["instance1", "instance2"])
  ami       = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}
```

27. How do you use loops in Terraform?

Loops in Terraform can be implemented using the `count` and `for_each` meta-arguments, as well as the `for` expression in variable assignments.

Example:

```
variable "instance_names" {
  type = list(string)
  default = ["instance1", "instance2"]
}

resource "aws_instance" "example" {
  for_each      = toset(var.instance_names)
  ami           = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}
```

28. What are locals in Terraform and how do you use them?

Locals in Terraform are used to define local values that can be reused within a module. They help avoid repetition and make configurations more readable.

Example:

```
locals {
  instance_type = "t2.micro"
  ami_id        = "ami-0c55b159cbfaffe1f0"
}

resource "aws_instance" "example" {
  ami           = local.ami_id
  instance_type = local.instance_type
}
```

29. What is the purpose of the `terraform taint` command?

`terraform taint` marks a resource for recreation on the next `terraform apply`. It is useful when a resource needs to be replaced due to a manual change or corruption.

Example:

```
terraform taint aws_instance.example
```

30. How do you manage module versioning in Terraform?

Module versioning in Terraform can be managed using the `version` argument in the `source` attribute of a module block, typically in combination with a registry.

Example:


```
module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  version = "2.0.0"  
}
```

31. What is the Terraform Registry?

The Terraform Registry is a public repository of Terraform modules and providers that can be used to discover and use pre-built modules and providers.

Example:

```
module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  version = "2.0.0"  
}
```

32. How do you perform a dry run in Terraform?

A dry run in Terraform can be performed using the `terraform plan` command, which shows the execution plan without making any changes.

Example:

```
terraform plan
```

33. What is the `terraform state` command used for?

The `terraform state` command is used to manage and manipulate the state file. It provides subcommands to move, remove, list, and inspect resources in the state file.

Example:

```
terraform state list
```

34. How do you rename a resource in the state file?

A resource can be renamed in the state file using the `terraform state mv` command, which moves the state of a resource to a new address.

Example:

```
terraform state mv aws_instance.old_name aws_instance.new_name
```

35. What is the purpose of the `terraform workspace` command?

The `terraform workspace` command is used to manage multiple workspaces, allowing for different states to be associated with the same configuration.

Example:

```
terraform workspace new dev
terraform workspace select dev
```

36. How do you debug Terraform configurations?

Debugging Terraform configurations can be done using the `TF_LOG` environment variable to set the log level and the `terraform console` command to interact with the configuration.

Example:

```
export TF_LOG=DEBUG
terraform apply
```

37. What is the difference between local and remote backends in Terraform?

Local backends store the state file on the local filesystem, while remote backends store the state file in a remote storage service (e.g., S3, Consul).

Example (local backend):

```
terraform {
  backend "local" {
    path = "terraform.tfstate"
  }
}
```

Example (remote backend):

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}
```

38. How do you handle provider versioning in Terraform?

Provider versioning in Terraform is managed using the `required_providers` block in the `terraform` block, specifying the version constraints.

Example:

```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 3.0"
    }
  }
}
```

39. What is the purpose of the `terraform refresh` command?

`terraform refresh` updates the state file with the current state of the infrastructure without making any changes to the configuration.

Example:

```
terraform refresh
```

40. How do you generate and view a resource graph in Terraform?

A resource graph can be generated using the `terraform graph` command and can be viewed using tools like Graphviz.

Example:

```
terraform graph | dot -Tpng > graph.png
```

41. What are `lifecycle` blocks in Terraform?

`lifecycle` blocks in Terraform are used to customize the lifecycle of a resource, such as creating before destroying, ignoring changes, and preventing deletion.

Example:

```
resource "aws_instance" "example" {
  ami           = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"

  lifecycle {
    create_before_destroy = true
  }
}
```

42. How do you ignore changes to a resource attribute in Terraform?

Changes to a resource attribute can be ignored using the `ignore_changes` argument in a `lifecycle` block.

Example:

```
resource "aws_instance" "example" {
  ami           = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"

  lifecycle {
    ignore_changes = [ami]
  }
}
```

43. What is the `terraform import` command used for?

`terraform import` is used to import existing infrastructure into Terraform's state file, mapping it to resources defined in the configuration.

Example:

```
terraform import aws_instance.example i-1234567890abcdef0
```

44. How do you use output values across different modules in Terraform?

Output values from one module can be referenced in another module by using the module's output attributes.

Example:

```
module "vpc" {  
    source = "../modules/vpc"  
}  
  
output "vpc_id" {  
    value = module.vpc.vpc_id  
}
```

45. What is the difference between `terraform output` and output values in configuration?

`terraform output` is a command that displays the output values of a Terraform configuration, while output values in configuration are defined using the `output` block.

Example (command):

```
terraform output
```

Example (configuration):

```
output "instance_id" {  
    value = aws_instance.example.id  
}
```

46. What are dynamic blocks in Terraform?

Dynamic blocks in Terraform are used to generate multiple nested blocks within a resource or module based on dynamic content.

Example:

```
resource "aws_security_group" "example" {  
    name      = "example-sg"  
    description = "Example security group"  
  
    dynamic "ingress" {  
        for_each = var.ingress_rules  
        content {  
            from_port    = ingress.value.from_port  
            to_port      = ingress.value.to_port  
            protocol      = ingress.value.protocol  
            cidr_blocks   = ingress.value.cidr_blocks  
        }  
    }  
}
```

```
}  
}
```

47. How do you define and use maps in Terraform?

Maps in Terraform are defined using the `map` type and can be used to store key-value pairs. They are accessed using the key.

Example:

```
variable "ami_ids" {  
  type = map(string)  
  default = {  
    us-east-1 = "ami-0c55b159cbfafa1f0"  
    us-west-2 = "ami-0d5eff06f840b45e9"  
  }  
}  
  
resource "aws_instance" "example" {  
  ami           = var.ami_ids[var.region]  
  instance_type = "t2.micro"  
}
```

48. What is a `count` parameter in Terraform?

The

`count` parameter in Terraform is used to create multiple instances of a resource based on a specified number.

Example:

```
resource "aws_instance" "example" {  
  count      = 3  
  ami       = "ami-0c55b159cbfafa1f0"  
  instance_type = "t2.micro"  
}
```

49. What are Terraform Cloud and Terraform Enterprise?

Terraform Cloud and Terraform Enterprise are commercial versions of Terraform that provide collaboration, governance, and automation features.

Example:

```
terraform {  
  backend "remote" {  
    organization = "my-org"  
    workspaces {  
      name = "my-workspace"  
    }  
  }  
}
```

50. How do you use a Terraform backend?

A backend is configured using the `terraform` block in the configuration file, specifying the backend type and its configuration.

Example:

```
terraform {  
  backend "s3" {  
    bucket = "my-terraform-state"  
    key     = "global/s3/terraform.tfstate"  
    region = "us-west-2"  
  }  
}
```

51. What is the `terraform fmt` command used for?

`terraform fmt` formats the configuration files to follow the Terraform style guide, making the code consistent and readable.

Example:

```
terraform fmt
```

52. How do you use a lock file in Terraform?

A lock file (`.terraform.lock.hcl`) is used to lock provider versions, ensuring consistency in provider versions across different environments.

Example:

```
terraform init
```

53. What is the purpose of the `terraform workspace` command?

The `terraform workspace` command is used to create, select, and manage multiple workspaces, allowing different states to be associated with the same configuration.

Example:

```
terraform workspace new dev  
terraform workspace select dev
```

54. How do you manage secrets in Terraform?

Secrets can be managed using environment variables, secure secret management services (e.g., AWS Secrets Manager), or Terraform's sensitive attribute.

Example:

```
resource "aws_secretsmanager_secret" "example" {
  name          = "example"
  description   = "An example secret"
}

resource "aws_secretsmanager_secret_version" "example" {
  secret_id     = aws_secretsmanager_secret.example.id
  secret_string = jsonencode({
    username = "example_user"
    password = "example_password"
  })
}
```

55. What is the `terraform console` command used for?

`terraform console` opens an interactive console for evaluating expressions, testing interpolation syntax, and debugging configurations.

Example:

```
terraform console
```

56. How do you reference data sources in Terraform?

Data sources are referenced using the `data` block and can be used to fetch information about existing infrastructure or services.

Example:

```
data "aws_ami" "example" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn-ami-hvm-*"]
  }
}

resource "aws_instance" "example" {
  ami           = data.aws_ami.example.id
  instance_type = "t2.micro"
}
```

57. What is the purpose of the `terraform state mv` command?

`terraform state mv` moves a resource in the state file to a new address, useful for renaming resources without recreating them.

Example:

```
terraform state mv aws_instance.old_name aws_instance.new_name
```

58. How do you use conditional expressions in Terraform?

Conditional expressions in Terraform are used to assign values based on conditions using the ternary operator `condition ? true_value : false_value`.

Example:

```
variable "environment" {
  default = "dev"
}

resource "aws_instance" "example" {
  ami           = "ami-0c55b159cbfaffe1f0"
  instance_type = var.environment == "prod" ? "t2.large" : "t2.micro"
}
```

59. What is the `terraform taint` command used for?

`terraform taint` marks a resource for recreation on the next `terraform apply`. It is useful when a resource needs to be replaced due to a manual change or corruption.

Example:

```
terraform taint aws_instance.example
```

60. How do you define and use maps in Terraform?

Maps in Terraform are defined using the `map` type and can be used to store key-value pairs. They are accessed using the key.

Example:

```
variable "ami_ids" {
  type = map(string)
  default = {
    us-east-1 = "ami-0c55b159cbfaffe1f0"
    us-west-2 = "ami-0d5eff06f840b45e9"
  }
}

resource "aws_instance" "example" {
  ami           = var.ami_ids[var.region]
  instance_type = "t2.micro"
}
```

61. How do you handle provider versioning in Terraform?

Provider versioning in Terraform is managed using the `required_providers` block in the `terraform` block, specifying the version constraints.

Example:

```
terraform {
  required_providers {
    aws = {
```



```

        source = "hashicorp/aws"
        version = "~> 3.0"
    }
}
}

```

62. What is the purpose of the `terraform refresh` command?

`terraform refresh` updates the state file with the current state of the infrastructure without making any changes to the configuration.

Example:

```
terraform refresh
```

63. What are `lifecycle` blocks in Terraform?

`lifecycle` blocks in Terraform are used to customize the lifecycle of a resource, such as creating before destroying, ignoring changes, and preventing deletion.

Example:

```

resource "aws_instance" "example" {
  ami           = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"

  lifecycle {
    create_before_destroy = true
  }
}

```

64. How do you use loops in Terraform?

Loops in Terraform can be implemented using the `count` and `for_each` meta-arguments, as well as the `for` expression in variable assignments.

Example:

```

variable "instance_names" {
  type = list(string)
  default = ["instance1", "instance2"]
}

resource "aws_instance" "example" {
  for_each      = toset(var.instance_names)
  ami           = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}

```

65. What is the `terraform import` command used for?

`terraform import` is used to import existing infrastructure into Terraform's state file, mapping it to resources defined in the configuration.

Example:

```
terraform import aws_instance.example i-1234567890abcdef0
```

66. How do you use output values across different modules in Terraform?

Output values from one module can be referenced in another module by using the module's output attributes.

Example:

```
module "vpc" {
  source = "../modules/vpc"
}

output "vpc_id" {
  value = module.vpc.vpc_id
}
```

67. What is the difference between `terraform output` and output values in configuration?

`terraform output` is a command that displays the output values of a Terraform configuration, while output values in configuration are defined using the `output` block.

Example (command):

```
terraform output
```

Example (configuration):

```
output "instance_id" {
  value = aws_instance.example.id
}
```

68. What are dynamic blocks in Terraform?

Dynamic blocks in Terraform are used to generate multiple nested blocks within a resource or module based on dynamic content.

Example:

```
resource "aws_security_group" "example" {
  name       = "example-sg"
  description = "Example security group"

  dynamic "ingress" {
    for_each = var.ingress_rules
    content {
      from_port = ingress.value.from_port
    }
  }
}
```

```

        to_port      = ingress.value.to_port
        protocol     = ingress.value.protocol
        cidr_blocks  = ingress.value.cidr_blocks
    }
}
}

```

69. How do you manage different environments (e.g., dev, prod) in Terraform?

Different environments can be managed using workspaces or separate directories with different variable files and state files.

Example:

```

terraform workspace new dev
terraform workspace new prod

```

70. How do you handle secrets in Terraform?

Secrets can be managed using environment variables, secure secret management services (e.g., AWS Secrets Manager), or Terraform's sensitive attribute.

Example:

```

resource "aws_secretsmanager_secret" "example" {
    name      = "example"
    description = "An example secret"
}

resource "aws_secretsmanager_secret_version" "example" {
    secret_id      = aws_secretsmanager_secret.example.id
    secret_string = jsonencode({
        username = "example_user"
        password = "example_password"
    })
}

```

71. What is the `terraform console` command used for?

`terraform console` opens an interactive console for evaluating expressions, testing interpolation syntax, and debugging

configurations.

Example:

```

terraform console

```

72. How do you reference data sources in Terraform?

Data sources are referenced using the `data` block and can be used to fetch information about existing infrastructure or services.

Example:

```
data "aws_ami" "example" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name     = "name"
    values   = ["amzn-ami-hvm-*"]
  }
}

resource "aws_instance" "example" {
  ami           = data.aws_ami.example.id
  instance_type = "t2.micro"
}
```

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Example:

```
terraform state mv aws_instance.old_name aws_instance.new_name
```

74. What is a backend in Terraform?

A backend in Terraform defines where and how state is loaded and stored. It can be local or remote (e.g., S3, Consul, etc.).

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}
```

75. How do you secure the state file in Terraform?

State files can be secured by storing them in remote backends with proper access controls and encryption, such as AWS S3 with server-side encryption and access control policies.

Example:

```
terraform {
  backend "s3" {
```

```

    bucket = "my-terraform-state"
    key     = "global/s3/terraform.tfstate"
    region = "us-west-2"
    encrypt = true
  }
}

```

76. What is the difference between local and remote backends in Terraform?

Local backends store the state file on the local filesystem, while remote backends store the state file in a remote storage service (e.g., S3, Consul).

Example (local backend):

```

terraform {
  backend "local" {
    path = "terraform.tfstate"
  }
}

```

Example (remote backend):

```

terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}

```

77. How do you manage module versioning in Terraform?

Module versioning in Terraform can be managed using the `version` argument in the `source` attribute of a module block, typically in combination with a registry.

Example:

```

module "vpc" {
  source = "terraform-aws-modules/vpc/aws"
  version = "2.0.0"
}

```

78. What is the Terraform Registry?

The Terraform Registry is a public repository of Terraform modules and providers that can be used to discover and use pre-built modules and providers.

Example:

```

module "vpc" {
  source = "terraform-aws-modules/vpc/aws"
  version = "2.0.0"
}

```

79. How do you generate and view a resource graph in Terraform?

A resource graph can be generated using the `terraform graph` command and can be viewed using tools like Graphviz.

Example:

```
terraform graph | dot -Tpng > graph.png
```

80. What is the purpose of the `terraform validate` command?

`terraform validate` is used to validate the syntax and configuration of the Terraform files without creating any resources.

Example:

```
terraform validate
```

81. What is the `terraform fmt` command used for?

`terraform fmt` formats the configuration files to follow the Terraform style guide, making the code consistent and readable.

Example:

```
terraform fmt
```

82. What are locals in Terraform and how do you use them?

Locals in Terraform are used to define local values that can be reused within a module. They help avoid repetition and make configurations more readable.

Example:

```
locals {
  instance_type = "t2.micro"
  ami_id        = "ami-0c55b159cbfafa1f0"
}

resource "aws_instance" "example" {
  ami           = local.ami_id
  instance_type = local.instance_type
}
```

83. How do you handle provider dependencies in Terraform?

Provider dependencies in Terraform are managed using the `required_providers` block in the `terraform` block, specifying the version constraints.

Example:

```

terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 3.0"
    }
  }
}

```

84. What is the `terraform apply` command used for?

`terraform apply` applies the changes required to reach the desired state of the configuration. It executes the plan created by `terraform plan`.

Example:

```
terraform apply
```

85. What is the `terraform destroy` command used for?

`terraform destroy` is used to destroy the infrastructure managed by Terraform. It removes all the resources defined in the configuration.

Example:

```
terraform destroy
```

86. What are output values in Terraform and how are they used?

Output values are used to extract information from the resources and make it accessible after the `apply` phase. They can be used to output resource attributes.

Example:

```

output "instance_id" {
  value = aws_instance.example.id
}

```

87. How do you manage different environments (e.g., dev, prod) in Terraform?

Different environments can be managed using workspaces or separate directories with different variable files and state files.

Example:

```

terraform workspace new dev
terraform workspace new prod

```

88. What is the difference between `count` and `for_each` in Terraform?

`count` is used to create multiple instances of a resource, while `for_each` is used to iterate over a map or set of values to create multiple instances.

Example (count):

```
resource "aws_instance" "example" {
  count      = 3
  ami       = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
}
```

Example (for_each):

```
resource "aws_instance" "example" {
  for_each   = toset(["instance1", "instance2"])
  ami       = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}
```

89. How do you use loops in Terraform?

Loops in Terraform can be implemented using the `count` and `for_each` meta-arguments, as well as the `for` expression in variable assignments.

Example:

```
variable "instance_names" {
  type = list(string)
  default = ["instance1", "instance2"]
}

resource "aws_instance" "example" {
  for_each   = toset(var.instance_names)
  ami       = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}
```

90. What is a `count` parameter in Terraform?

The `count` parameter in Terraform is used to create multiple instances of a resource based on a specified number.

Example:

```
resource "aws_instance" "example" {
  count      = 3
  ami       = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
}
```


91. What are Terraform Cloud and Terraform Enterprise?

Terraform Cloud and Terraform Enterprise are commercial versions of Terraform that provide collaboration, governance, and automation features.

Example:

```
terraform {
  backend "remote" {
    organization = "my-org"
    workspaces {
      name = "my-workspace"
    }
  }
}
```

92. How do you use a Terraform backend?

A backend is configured using the `terraform` block in the configuration file, specifying the backend type and its configuration.

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}
```

93. What is the `terraform fmt` command used for?

`terraform fmt` formats the configuration files to follow the Terraform style guide, making the code consistent and readable.

Example:

```
terraform fmt
```

94. How do you use a lock file in Terraform?

A lock file (`.terraform.lock.hcl`) is used to lock provider versions, ensuring consistency in provider versions across different environments.

Example:

```
terraform init
```

95. What is the purpose of the `terraform workspace` command?

The `terraform workspace` command is used to create, select, and manage multiple workspaces, allowing different states to be associated with the same configuration.

Example:

```
terraform workspace new dev
terraform workspace select dev
```

96. ****How do you manage secrets**

in Terraform?**

Secrets can be managed using environment variables, secure secret management services (e.g., AWS Secrets Manager), or Terraform's sensitive attribute.

Example:

```
resource "aws_secretsmanager_secret" "example" {
  name       = "example"
  description = "An example secret"
}

resource "aws_secretsmanager_secret_version" "example" {
  secret_id      = aws_secretsmanager_secret.example.id
  secret_string = jsonencode({
    username = "example_user"
    password = "example_password"
  })
}
```

97. What is the `terraform console` command used for?

`terraform console` opens an interactive console for evaluating expressions, testing interpolation syntax, and debugging configurations.

Example:

```
terraform console
```

98. How do you reference data sources in Terraform?

Data sources are referenced using the `data` block and can be used to fetch information about existing infrastructure or services.

Example:

```
data "aws_ami" "example" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn-ami-hvm-*"]
  }
}
```

```

    }
}

resource "aws_instance" "example" {
  ami           = data.aws_ami.example.id
  instance_type = "t2.micro"
}

```

99. What is the purpose of the `terraform state mv` command?

`terraform state mv` moves a resource in the state file to a new address, useful for renaming resources without recreating them.

Example:

```
terraform state mv aws_instance.old_name aws_instance.new_name
```

100. What is a backend in Terraform?

A backend in Terraform defines where and how state is loaded and stored. It can be local or remote (e.g., S3, Consul, etc.).

Example:

```

terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}

```

101. How do you secure the state file in Terraform?

State files can be secured by storing them in remote backends with proper access controls and encryption, such as AWS S3 with server-side encryption and access control policies.

Example:

```

terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
    encrypt = true
  }
}

```

102. What is the difference between local and remote backends in Terraform?

Local backends store the state file on the local filesystem, while remote backends store the state file in a remote storage service (e.g., S3, Consul).

Example (local backend):

```
terraform {
  backend "local" {
    path = "terraform.tfstate"
  }
}
```

Example (remote backend):

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}
```

103. How do you manage module versioning in Terraform?

Module versioning in Terraform can be managed using the `version` argument in the `source` attribute of a module block, typically in combination with a registry.

Example:

```
module "vpc" {
  source = "terraform-aws-modules/vpc/aws"
  version = "2.0.0"
}
```

104. What is the Terraform Registry?

The Terraform Registry is a public repository of Terraform modules and providers that can be used to discover and use pre-built modules and providers.

Example:

```
module "vpc" {
  source = "terraform-aws-modules/vpc/aws"
  version = "2.0.0"
}
```

105. How do you generate and view a resource graph in Terraform?

A resource graph can be generated using the `terraform graph` command and can be viewed using tools like Graphviz.

Example:

```
terraform graph | dot -Tpng > graph.png
```

106. What is the purpose of the `terraform validate` command?

`terraform validate` is used to validate the syntax and configuration of the Terraform files without creating any resources.

Example:

```
terraform validate
```

107. What is the `terraform fmt` command used for?

`terraform fmt` formats the configuration files to follow the Terraform style guide, making the code consistent and readable.

Example:

```
terraform fmt
```

108. What are locals in Terraform and how do you use them?

Locals in Terraform are used to define local values that can be reused within a module. They help avoid repetition and make configurations more readable.

Example:

```
locals {
  instance_type = "t2.micro"
  ami_id        = "ami-0c55b159cbfafa1f0"
}

resource "aws_instance" "example" {
  ami           = local.ami_id
  instance_type = local.instance_type
}
```

109. How do you handle provider dependencies in Terraform?

Provider dependencies in Terraform are managed using the `required_providers` block in the `terraform` block, specifying the version constraints.

Example:

```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 3.0"
    }
  }
}
```

110. What is the `terraform apply` command used for?

`terraform apply` applies the changes required to reach the desired state of the configuration. It executes the plan created by `terraform plan`.

Example:

```
terraform apply
```

111. What is the `terraform destroy` command used for?

`terraform destroy` is used to destroy the infrastructure managed by Terraform. It removes all the resources defined in the configuration.

Example:

```
terraform destroy
```

112. What are output values in Terraform and how are they used?

Output values are used to extract information from the resources and make it accessible after the `apply` phase. They can be used to output resource attributes.

Example:

```
output "instance_id" {
  value = aws_instance.example.id
}
```

113. How do you manage different environments (e.g., dev, prod) in Terraform?

Different environments can be managed using workspaces or separate directories with different variable files and state files.

Example:

```
terraform workspace new dev
terraform workspace new prod
```

114. What is the difference between `count` and `for_each` in Terraform?

`count` is used to create multiple instances of a resource, while `for_each` is used to iterate over a map or set of values to create multiple instances.

Example (count):

```
resource "aws_instance" "example" {
  count      = 3
  ami       = "ami-0c55b159cbfafa1f0"
  instance_type = "t2.micro"
}
```

Example (for_each):

```
resource "aws_instance" "example" {
  for_each      = toset(["instance1", "instance2"])
  ami           = "ami-0c55b159cbfafa1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}
```

115. How do you use loops in Terraform?

Loops in Terraform can be implemented using the `count` and `for_each` meta-arguments, as well as the `for` expression in variable assignments.

Example:

```
variable "instance_names" {
  type = list(string)
  default = ["instance1", "instance2"]
}

resource "aws_instance" "example" {
  for_each      = toset(var.instance_names)
  ami           = "ami-0c55b159cbfafa1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}
```

116. What is a `count` parameter in Terraform?

The `count` parameter in Terraform is used to create multiple instances of a resource based on a specified number.

Example:

```
resource "aws_instance" "example" {
  count      = 3
  ami       = "ami-0c55b159cbfafa1f0"
  instance_type = "t2.micro"
}
```

117. What are Terraform Cloud and Terraform Enterprise?

Terraform Cloud and Terraform Enterprise are commercial versions of Terraform that provide collaboration, governance, and automation features.

Example:

```
terraform {
  backend "remote" {
```

```

    organization = "my-org"
    workspaces {
      name = "my-workspace"
    }
  }
}

```

118. How do you use a Terraform backend?

A backend is configured using the `terraform` block in the configuration file, specifying the backend type and its configuration.

Example:

```

terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}

```

119. What is the `terraform fmt` command used for?

`terraform fmt` formats the configuration files

to follow the Terraform style guide, making the code consistent and readable.

Example:

```
terraform fmt
```

120. How do you use a lock file in Terraform?

A lock file (`.terraform.lock.hcl`) is used to lock provider versions, ensuring consistency in provider versions across different environments.

Example:

```
terraform init
```

121. What is the purpose of the `terraform workspace` command?

The `terraform workspace` command is used to create, select, and manage multiple workspaces, allowing different states to be associated with the same configuration.

Example:

```

terraform workspace new dev
terraform workspace select dev

```


122. How do you manage secrets in Terraform?

Secrets can be managed using environment variables, secure secret management services (e.g., AWS Secrets Manager), or Terraform's sensitive attribute.

Example:

```
resource "aws_secretsmanager_secret" "example" {
  name          = "example"
  description   = "An example secret"
}

resource "aws_secretsmanager_secret_version" "example" {
  secret_id      = aws_secretsmanager_secret.example.id
  secret_string = jsonencode({
    username = "example_user"
    password = "example_password"
  })
}
```

123. What is the `terraform console` command used for?

`terraform console` opens an interactive console for evaluating expressions, testing interpolation syntax, and debugging configurations.

Example:

```
terraform console
```

124. How do you reference data sources in Terraform?

Data sources are referenced using the `data` block and can be used to fetch information about existing infrastructure or services.

Example:

```
data "aws_ami" "example" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn-ami-hvm-*"]
  }
}

resource "aws_instance" "example" {
  ami           = data.aws_ami.example.id
  instance_type = "t2.micro"
}
```

125. What is the purpose of the `terraform state mv` command?

`terraform state mv` moves a resource in the state file to a new address, useful for renaming resources without recreating them.

Example:

```
terraform state mv aws_instance.old_name aws_instance.new_name
```

126. What is a backend in Terraform?

A backend in Terraform defines where and how state is loaded and stored. It can be local or remote (e.g., S3, Consul, etc.).

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}
```

127. How do you secure the state file in Terraform?

State files can be secured by storing them in remote backends with proper access controls and encryption, such as AWS S3 with server-side encryption and access control policies.

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
    encrypt = true
  }
}
```

128. What is the difference between local and remote backends in Terraform?

Local backends store the state file on the local filesystem, while remote backends store the state file in a remote storage service (e.g., S3, Consul).

Example (local backend):

```
terraform {
  backend "local" {
    path = "terraform.tfstate"
  }
}
```

Example (remote backend):

```

terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key     = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}

```

129. How do you manage module versioning in Terraform?

Module versioning in Terraform can be managed using the `version` argument in the `source` attribute of a module block, typically in combination with a registry.

Example:

```

module "vpc" {
  source = "terraform-aws-modules/vpc/aws"
  version = "2.0.0"
}

```

130. What is the Terraform Registry?

The Terraform Registry is a public repository of Terraform modules and providers that can be used to discover and use pre-built modules and providers.

Example:

```

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```

terraform validate

```

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Example:

```
terraform fmt
```

134. What are locals in Terraform and how do you use them?

Locals in Terraform are used to define local values that can be reused within a module. They help avoid repetition and make configurations more readable.

Example:

```
locals {
  instance_type = "t2.micro"
  ami_id        = "ami-0c55b159cbfaffe1f0"
}

resource "aws_instance" "example" {
  ami           = local.ami_id
  instance_type = local.instance_type
}
```

135. How do you handle provider dependencies in Terraform?

Provider dependencies in Terraform are managed using the `required_providers` block in the `terraform` block, specifying the version constraints.

Example:

```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 3.0"
    }
  }
}
```

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Example (count):

```
resource "aws_instance" "example" {
  count      = 3
  ami       = "ami-0c55b159cbf1f0"
  instance_type = "t2.micro"
}
```

Example (for_each):

```
resource "aws_instance" "example" {
  for_each   = toset(["instance1", "instance2"])
  ami       = "ami-0c55b159cbf1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
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      name = "my-workspace"
    }
  }
}
```

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A backend is configured using the `terraform` block in the configuration file, specifying the backend type and its configuration.

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
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    region = "us-west-2"
  }
}
```

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```

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Example:

```
terraform workspace new dev
terraform workspace select dev
```

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Example:

```
resource "aws_secretsmanager_secret" "example" {
  name          = "example"
  description   = "An example secret"
}

resource "aws_secretsmanager_secret_version" "example" {
  secret_id     = aws_secretsmanager_secret.example.id
  secret_string = jsonencode({
    username = "example_user"
    password = "example_password"
  })
}
```

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```

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  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn-ami-hvm-*"]
  }
}
```

```
resource "aws_instance" "example" {
  ami          = data.aws_ami.example.id
  instance_type = "t2.micro"
}
```

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Example:

```
terraform state mv aws_instance.old_name aws_instance.new_name
```

152. What is a backend in Terraform?

A backend in Terraform defines where and how state is loaded and stored. It can be local or remote (e.g., S3, Consul, etc.).

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}
```

153. How do you secure the state file in Terraform?

State files can be secured by storing them in remote backends with proper access controls and encryption, such as AWS S3 with server-side encryption and access control policies.

Example:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
    encrypt = true
  }
}
```

154. What is the difference between local and remote backends in Terraform?

Local backends store the state file on the local filesystem, while remote backends store the state file in a remote storage service (e.g., S3, Consul).

Example (local backend):

```
terraform {  
  backend "local" {  
    path = "terraform.tfstate"  
  }  
}
```

Example (remote backend):

```
terraform {  
  backend "s3" {  
    bucket = "my-terraform-state"  
    key    = "global/s3/terraform.tfstate"  
    region = "us-west-2"  
  }  
}
```

155. How do you manage module versioning in Terraform?

Module versioning in Terraform can be managed using the `version` argument in the `source` attribute of a module block, typically in combination with a registry.

Example:

```
module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  version = "2.0.0"  
}
```

156. What is the Terraform Registry?

The Terraform Registry is a public repository of Terraform modules and providers that can be used to discover and use pre-built modules and providers.

Example:

```
module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  version = "2.0.0"  
}
```

157. How do you generate and view a resource graph in Terraform?

A resource graph can be generated using the `terraform graph` command and can be viewed using tools like Graphviz.

Example:

```
terraform graph | dot -Tpng > graph.png
```

158. What is the purpose of the `terraform validate` command?

`terraform validate` is used to validate the syntax and configuration of the Terraform files without creating any resources.

Example:

```
terraform validate
```

159. What is the `terraform fmt` command used for?

`terraform fmt` formats the configuration files to follow the Terraform style guide, making the code consistent and readable.

Example:

```
terraform fmt
```

160. What are locals in Terraform and how do you use them?

Locals in Terraform are used to define local values that can be reused within a module. They help avoid repetition and make configurations more readable.

Example:

```
locals {
  instance_type = "t2.micro"
  ami_id        = "ami-0c55b159cbfafa1f0"
}

resource "aws_instance" "example" {
  ami           = local.ami_id
  instance_type = local.instance_type
}
```

161. How do you handle provider dependencies in Terraform?

Provider dependencies in Terraform are managed using the `required_providers` block in the `terraform` block, specifying the version constraints.

Example:

```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 3.0"
    }
  }
}
```

162. What is the `terraform apply` command used for?

`terraform apply` applies the changes required to reach the desired state of the configuration. It executes the plan created by `terraform plan`.

Example:

```
terraform apply
```

163. What is the `terraform destroy` command used for?

`terraform destroy` is used to destroy the infrastructure managed by Terraform. It removes all the resources defined in the configuration.

Example:

```
terraform destroy
```

164. What are output values in Terraform and how are they used?

Output values are used to extract information from the resources and make it accessible after the `apply` phase. They can be used to output resource attributes.

Example:

```
output "instance_id" {
  value = aws_instance.example.id
}
```

165. How do you manage different environments (e.g., dev, prod) in Terraform?

Different environments can be managed using workspaces or separate directories with different variable files and state files.

Example:

```
terraform workspace new dev
terraform workspace new prod
```

166. What is the difference between `count` and `for_each` in Terraform?

`count` is used to create multiple instances of a resource, while `for_each` is used to iterate over a map or set of values to create multiple instances.

Example (count):

```
resource "aws_instance" "example" {
  count      = 3
  ami       = "ami-0c55b159cbfafa1f0"
  instance_type = "t2.micro"
}
```

Example (for_each):

```
resource "aws_instance" "example" {
  for_each      = toset(["instance1", "instance2"])
  ami           = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}
```

167. How do you use loops in Terraform?

Loops in Terraform can be implemented using the `count` and `for_each` meta-arguments, as well as the `for` expression in variable assignments.

Example:

```
variable "instance_names" {
  type = list(string)
  default = ["instance1", "instance2"]
}

resource "aws_instance" "example" {
  for_each      = toset(var.instance_names)
  ami           = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}
```

168. What is a `count` parameter in Terraform?

The `count` parameter in Terraform is used to create multiple instances of a resource based on a specified number.

Example:

```
resource "aws_instance" "example" {
  count      = 3
  ami       = "ami-0c55b159cbfaffe1f0"
  instance_type = "t2.micro"
}
```

169. What are Terraform Cloud and Terraform Enterprise?

Terraform Cloud and Terraform Enterprise are commercial versions of Terraform that provide collaboration, governance, and automation features.

Example:

```

terraform {
  backend "remote" {
    organization = "my-org"
    workspaces {
      name = "my-workspace"
    }
  }
}

```

170. How do you use a Terraform backend?

A backend is configured using the `terraform` block in the configuration file, specifying the backend type and its configuration.

Example:

```

terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "global/s3/terraform.tfstate"
    region = "us-west-2"
  }
}

```

171. What is the `terraform fmt` command used for?

`terraform fmt` formats the configuration files to follow the Terraform style guide, making the code consistent and readable.

Example:

```

terraform fmt

```

172. How do you use a lock file in Terraform?

A lock file (`.terraform.lock.hcl`) is used to lock provider versions, ensuring consistency in provider versions across different environments.

Example:

```

terraform init

```

173. What is the purpose of the `terraform workspace` command?

The `terraform workspace` command is used to create, select, and manage multiple workspaces, allowing different states to be associated with the same configuration.

Example:

```

terraform workspace new dev
terraform workspace select dev

```

174. How do you manage secrets in Terraform?

Secrets can be managed using environment variables, secure secret management services (e.g., AWS Secrets Manager), or Terraform's sensitive attribute.

Example:

```
resource "aws_secretsmanager_secret" "example" {
  name          = "example"
  description   = "An example secret"
}

resource "aws_secretsmanager_secret_version" "example" {
  secret_id     = aws_secretsmanager_secret.example.id
  secret_string = jsonencode({
    username = "example_user"
    password = "example_password"
  })
}
```

175. What is the `terraform console` command used for?

`terraform console` opens an interactive console for evaluating expressions, testing interpolation syntax, and debugging configurations.

Example:

```
terraform console
```

176. How do you reference data sources in Terraform?

Data sources are referenced using the `data` block and can be used to fetch information about existing infrastructure or services.

Example:

```
data "aws_ami" "example" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn-ami-hvm-*"]
  }
}

resource "aws_instance" "example" {
  ami           = data.aws_ami.example.id
  instance_type = "t2.micro"
}
```

177. What is the purpose of the `terraform state mv` command?

`terraform state mv` moves a resource in the state file to a new address, useful for renaming resources without recreating them.

Example:

```
terraform state mv aws_instance.old_name aws_instance.new_name
```

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```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
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Example (local backend):

```
terraform {
  backend "local" {
    path = "terraform.tfstate"
  }
}
```

Example (remote backend):

```

terraform {
  backend "s3" {
    bucket = "my-terraform-state"
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    region = "us-west-2"
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```

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```
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  ami_id        = "ami-0c55b159cbfaffe1f0"
}

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  instance_type = local.instance_type
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  instance_type = "t2.micro"
}
```

Example (for

_each):

```
resource "aws_instance" "example" {
  for_each   = toset(["instance1", "instance2"])
  ami       = "ami-0c55b159cbf1f0"
  instance_type = "t2.micro"
  tags = {
    Name = each.key
  }
}
```

```
}
```

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}
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terraform init
```

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```
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terraform workspace select dev
```

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Example:

```
resource "aws_secretsmanager_secret" "example" {  
  name       = "example"  
  description = "An example secret"
```

```
}  
  
resource "aws_secretsmanager_secret_version" "example" {  
  secret_id      = aws_secretsmanager_secret.example.id  
  secret_string = jsonencode({  
    username = "example_user"  
    password = "example_password"  
  })  
}
```